
Decontextualization, Everywhere: A Systematic Audit on PeerQA

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Abstract

We audit decontextualization strategies for long-document scientific QA on PeerQA. We sweep sentence- and paragraph-level templates (from minimal content to title+heading) across BM25, TF-IDF, dense retrieval, ColBERT, and cross-encoder reranking, and evaluate with Recall@k, MRR, and answerability F1. A central finding is that oracle-style evaluation (per-paper indexing) dramatically inflates retrieval scores compared to full-corpus search: BM25 achieves $R@10=1.000$ and $MRR\approx 0.68$ under oracle, but only $R@10\approx 0.011$ and $MRR\approx 0.015$ over the full corpus. Surprisingly, answerability remains robust, with full-corpus configurations matching or exceeding oracle F1. We further show that decontextualization is not one-size-fits-all: sparse methods favor minimal context in oracle settings, while paragraph-level chunks with measured structure (title+heading) work best under realistic full-corpus conditions, and late-interaction models benefit from more aggressive context. We release a configurable framework and provide practical guidance: prioritize paper identification before fine-grained evidence search, prefer paragraph-level chunks, use measured decontextualization, and evaluate end-to-end under full-corpus conditions.

1 Introduction

Scientific articles are long, structured documents in which the information relevant to a question is often sparse, non-contiguous, and phrased with domain-specific terminology [26, 24]. Building reliable question answering (QA) systems over such documents therefore hinges on effective retrieval of fine-grained evidence before any downstream inference [6, 21]. PeerQA [4] is a realistic benchmark for this setting: questions are sourced from peer reviews, answers are provided by authors, and sentence- and paragraph-level evidence is explicitly annotated [17, 24]. A central, recurring observation in this domain is that decontextualization—augmenting passages with structural cues such as the paper title or the most recent section heading—can improve retrieval [31, 12]. Yet, despite its growing use, we lack a systematic understanding of when, how, and to what extent decontextualization helps across retrieval families and how these choices propagate to downstream tasks.

We define decontextualization as the controlled addition of document structure to a target unit (sentence or paragraph) prior to indexing and retrieval. While adding context may help disambiguate

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short spans, it may also introduce lexical drift or bias similarity measures, particularly for sparse methods [31, 12]. Moreover, different retrieval architectures (sparse lexical, dense, late-interaction, and cross-encoder reranking) likely respond differently to such augmentation [29, 18, 19], and the optimal strategy may depend on the chunk granularity [24, 20]. Finally, improvements in retrieval do not always translate linearly to downstream performance in answerability classification or answer generation, raising the need for an end-to-end audit [21, 18].

In this work, we present a systematic, controlled audit of decontextualization on PeerQA. We sweep decontextualization templates that range from minimal (content only) to full (title + heading + content) at both sentence- and paragraph-level granularities. We evaluate four first-stage retrievers—BM25 and TF-IDF (sparse), a sentence-transformer dense retriever, and ColBERT (late interaction)—and a cross-encoder reranker [38, 41, 36, 18, 19, 30]. Retrieval is assessed with Recall@k and MRR [5, 15]. To quantify downstream impact, we propagate retrieval outputs into answerability classification and report F1.

Our contributions are as follows:

- A systematic audit of decontextualization across retrieval families (sparse, dense, late interaction) and granularities (sentence, paragraph) on PeerQA with author-verified evidence.
- Evidence that oracle evaluation substantially overstates retrieval effectiveness versus full-corpus search, explaining prior high scores and clarifying evaluation regimes.
- A characterization of retrieval–downstream decoupling: answerability F1 exhibits only weak correlation with retrieval quality, motivating evaluation beyond retrieval metrics.
- Practical guidance for system design: prioritize paper identification before fine-grained evidence search, prefer paragraph-level chunks, and apply measured decontextualization (title+heading) under full-corpus settings, with context tailored to retriever family.
- A configurable framework that sweeps templates and granularities, builds indexes, and reports retrieval (Recall@k, MRR) and answerability (F1) in both oracle and full-corpus regimes.

In preview, we show that oracle evaluation dramatically overstates retrieval compared to full-corpus search and that answerability F1 is only weakly coupled to retrieval quality. These findings lead to concrete guidance: prioritize paper identification, prefer paragraph-level chunks with measured decontextualization, and evaluate end-to-end under full-corpus conditions.

2 Related Work

Long-Document Scientific QA. Long-document scientific QA foregrounds evidence retrieval and domain grounding: QASPER pairs information-seeking questions with author-annotated answers and rationale spans in research papers [9]; QuALITY targets long-context reasoning [33]; PubMedQA and BioASQ emphasize biomedical QA with specialized terminology [16, 3]; S2ORC enables large-scale scholarly text experimentation [26]; and PeerRead highlights review discourse where local spans and structural cues matter for retrieval [17].

Retrieval Architectures and Re-ranking for QA. BM25 remains a strong sparse baseline, standardized by Anserini/Pyserini [38, 27, 23]; dense retrieval with dual encoders (e.g., DPR) and unsupervised variants like Contriever are staples across domains [18, 14]; late-interaction models (ColBERT, ColBERTv2) balance effectiveness and efficiency [19, 40]; and cross-encoder re-rankers (BERT, MonoT5) markedly improve ranking [30, 32]. BEIR shows method gains are dataset- and domain-specific [42].

Decontextualization and Structural Cues in Retrieval. Decontextualization via titles/headers has long benefited QA: DrQA and DPR concatenate titles, aiding disambiguation, while sentence-level settings (e.g., FEVER) use titles to keep evidence interpretable [6, 18, 43]. TREC CAR underscores complementary value from hierarchical structure [8]. Yet the strength of decontextualization depends on retriever family, domain, and chunk size—factors we audit in PeerQA-style pipelines across BM25, dense encoders, late-interaction models, and cross-encoder re-rankers.

Granularity: Sentence vs. Paragraph. Chunk size trades precision for context: paragraphs offer richer signals but can add distractors, whereas sentences pinpoint evidence but may lack disambiguating context [18, 2, 43]. Scientific QA (e.g., QASPER) surfaces this tension [9]; we study how title/heading decontextualization interacts with granularity and retriever family to mitigate context loss or distractor bias.

Retrieval-Augmented Generation and Answerability. RAG and FiD improve grounding by conditioning on retrieved evidence [21, 13], but retrieval choices can affect faithfulness, with structural cues sometimes biasing generation toward topical yet non-evidential content [28]. Unanswerability detection (SQuAD 2.0) offers safeguards [35]; evaluation spans ROUGE and QA/LLM-judge metrics for faithfulness and alignment [22, 11, 25]. We propagate retrieval variations from decontextualization to answerability and generation quality in scientific QA.

Overall, prior work shows that retrieval/re-ranking design, decontextualization via structural cues, and chunk granularity jointly shape effectiveness; we operationalize these insights in a controlled audit over scientific peer-review QA to provide dataset-native guidance on decontextualization across retriever families and granularities. We provide guidance on decontextualization best practices across retriever families and granularities.

3 Methods

3.1 Dataset and Experimental Setup

We conduct our experiments on the PeerQA dataset, which contains scientific questions derived from peer reviews with author-provided answers and evidence mappings. The dataset includes:

- QA pairs with question_id, question text, answer evidence, and answerability labels
- Extracted paper text with hierarchical structure (title, headings, paragraphs, sentences)
- Ground truth relevance judgments (qrels) at sentence and paragraph levels

Our experimental framework processes data at two granularities: sentence-level and paragraph-level chunking. For each granularity, we apply multiple decontextualization templates ranging from minimal (content only) to comprehensive (title + heading + content), motivated by prior work on making spans standalone and self-contained [7].

3.2 Decontextualization Templates

We design and evaluate four primary decontextualization templates:

1. **Minimal:** Raw content without additional context
2. **Title+Content:** “Title: {title} Content: {content}”
3. **Heading+Content:** “Heading: {last_heading} Content: {content}”
4. **Full Context:** “Title: {title} Heading: {last_heading} Content: {content}”

These templates are applied systematically across both sentence and paragraph granularities, creating a comprehensive evaluation matrix, and are aligned with prior approaches to decontextualizing spans for retrieval and reading tasks [7].

3.3 Retrieval Methods

We compare five retrieval approaches: BM25 and TF-IDF (sparse), a sentence-transformer dense retriever (all-MiniLM-L6-v2), ColBERT (late interaction), and a cross-encoder reranker. BM25 uses standard settings ($k_1=1.2$, $b=0.75$) [38]. TF-IDF follows the classic vector space model formulation [39]. Dense encoders use cosine similarity with optional FAISS for ANN search [36, 45, 10]. ColBERT applies MaxSim over token representations (late interaction) [19]. The cross-encoder reranks top-k candidates from a first-stage retriever using a BERT-style cross-encoder [30]. For sparse learned baselines referenced in comparisons (e.g., SPLADE), we follow prior work on expansion-based sparse retrieval [12].

3.4 Evaluation Methodology

3.4.1 Retrieval Evaluation

For each retriever and decontextualization configuration, we report Recall@k ($k \in \{1, 5, 10, 20, 50\}$) and MRR, standard IR metrics [5]. Relevance is taken from PeerQA’s author-provided evidence mappings at sentence and paragraph levels, in line with evidence-grounded evaluation used in scientific QA [44, 43].

3.4.2 Downstream Evaluation

We propagate retrieved contexts to answerability classification (binary F1) to measure how retrieval variations influence downstream decision-making. Answerability detection follows established practice from unanswerable-question benchmarks (e.g., SQuAD 2.0) [35].

3.5 Implementation Framework

We provide a configurable framework that sweeps templates and granularities, builds indexes, evaluates retrievers, and runs downstream tasks:

Algorithm 1 Decontextualization Audit Framework

```
1: Load PeerQA dataset (QA, papers, qrels)
2: for each granularity  $g \in \{\text{sentence, paragraph}\}$  do
3:   for each template  $t \in \text{Templates}$  do
4:     Apply decontextualization template  $t$  to documents at granularity  $g$ 
5:     for each retriever  $r \in \text{Retrievers}$  do
6:       Build index for  $r$  on processed documents
7:       Evaluate retrieval on test queries
8:       Record Recall@k and MRR
9:     end for
10:    Run downstream tasks using retrieval results
11:    Record answerability and generation metrics
12:  end for
13: end for
14: Analyze results across configurations
15: Generate comparative report and recommendations
```

The framework supports:

- Configurable retrieval methods with automatic dependency detection
- Batch processing for efficient evaluation
- Comprehensive metric collection and automated aggregation/reporting

4 Experimental Results

We conducted comprehensive experiments across 579 real Q&A pairs from 90 scientific papers, evaluating multiple retrieval methods with 5 decontextualization templates at 2 granularities. To understand the impact of search space on retrieval performance, we evaluated two distinct experimental settings: (1) Oracle evaluation with per-paper indexes, and (2) Full corpus evaluation across all documents.

4.1 Experimental Setup: Oracle vs. Full Corpus

A critical methodological consideration in evaluating retrieval systems is the search space size. In scientific QA, many evaluations operate in an *oracle* or within-document regime (e.g., QASPER), which dramatically simplifies retrieval by assuming the target paper is known a priori [44]. By contrast, open-domain settings require searching across many documents and are substantially more challenging [6, 21, 42, 34]:

- **Oracle Setting:** Creates separate indexes for each paper (averaging 270 chunks per paper). Questions are searched only within their source paper’s index, representing an idealized scenario where the relevant paper is known a priori [44].
- **Full Corpus Setting:** Creates a single index containing all 24,265 chunks from 90 papers. Questions must be retrieved from this entire collection, representing the realistic challenge of open-domain scientific QA [6, 42].

4.2 Comparison with Prior Baselines

We contrast our oracle-style setup with full-corpus retrieval to highlight the impact of search space. Oracle-style evaluation is common in scientific QA (e.g., QASPER) [44], whereas open-domain retrieval reflects realistic deployment conditions [42, 34]. The following tables summarize: (i) oracle retrieval performance for representative models, and (ii) best answerability classification scores contrasting oracle-style per-paper retrieval against our full-corpus setting. Note that prior work often reports macro-F1 for answerability due to class imbalance (cf. SQuAD 2.0’s emphasis on unanswerability) [35], whereas our downstream tables report overall F1.

Table 1: Oracle retrieval performance with per-paper indexes (270 chunks per paper). Paragraph-level results for representative models; “+Title” indicates decontextualization by prepending the paper title.

Model	MRR (Para.)	MRR (+Title)	R@10 (Para.)	R@10 (+Title)
BM25	0.4288	–	0.6388	–
ColBERTv2	0.4368	0.4122	0.6287	0.6371
SPLADEv3	0.4536	0.4725	0.6661	0.6851
BM25 (Ours, oracle)	0.679	0.680	1.000	1.000
BM25 (Ours, full corpus)	0.015	–	0.011	–
ColBERT (Ours, full corpus)	–	0.029	–	0.025

Table 2: Answerability classification: Oracle vs. Full Corpus (best scores). Prior oracle-style evaluations often employ strong LMs (e.g., GPT-4) [1]; ours uses retrieved contexts from the specified retrievers.

Setting	Metric	Best Score	Model/Config	Context
Oracle-style	Macro-F1	0.571	GPT-4	Top-50 passages
Ours (oracle)	F1	0.713	BM25 (para/aggressive_title)	Per-paper passages
Ours (full corpus)	F1	0.718	Dense (para/title_heading)	Retrieved passages

Comparison. Under oracle conditions, paragraph-level sparse/lexical and re-weighted sparse models (BM25, SPLADE) typically achieve strong MRR and recall [38, 12]. Our own BM25 oracle setting reaches R@10=1.000 and MRR=0.680 (para/aggressive_title), confirming the effect of drastically reduced search space. In full-corpus search, our best ColBERT configuration attains only R@10=0.025 and MRR=0.029, consistent with the increased difficulty of open-domain retrieval [42, 34]. Despite this large gap in retrieval, our best answerability score (F1=0.718) is competitive with oracle-style results, echoing findings that strong language models can make reliable unanswerability judgments even with limited or noisy context [35, 37].

4.3 Oracle Evaluation Results

Table 3 presents retrieval performance under oracle conditions, where search is restricted to the source paper of each question. These results align with prior within-document evaluations in scientific QA [44].

Under oracle conditions, BM25 achieves remarkably high performance, with paragraph-level retrieval reaching perfect Recall@10 (1.000) and strong MRR (0.680). This is consistent with the effectiveness of lexical matching when the search space is constrained [38].

Key observations from oracle evaluation:

- **Paragraph superiority:** Paragraph-level chunking dramatically outperforms sentence-level (Recall@10: 1.000 vs. 0.774), suggesting that paragraph boundaries better align with

Table 3: Oracle retrieval performance with per-paper indexes (270 chunks per paper)

Granularity	Template	Recall@5	Recall@10	Recall@20	MRR
<i>Sentence-level</i>					
Sentence	minimal	0.632	0.774	0.891	0.474
Sentence	title_only	0.629	0.771	0.889	0.473
Sentence	heading_only	0.630	0.775	0.891	0.473
Sentence	title_heading	0.627	0.769	0.887	0.472
Sentence	aggressive_title	0.632	0.770	0.889	0.474
<i>Paragraph-level</i>					
Paragraph	minimal	0.994	1.000	1.000	0.679
Paragraph	title_only	0.925	0.994	1.000	0.553
Paragraph	heading_only	0.938	0.994	1.000	0.567
Paragraph	title_heading	0.916	0.994	1.000	0.545
Paragraph	aggressive_title	0.994	1.000	1.000	0.680

evidence units in scientific text, in line with document-level QA settings where answers span multiple sentences [44, 46].

- **Minimal decontextualization optimal:** Unlike our hypothesis, minimal templates achieve the best performance in oracle settings, indicating that when searching within a single paper, additional context can introduce noise [5].
- **Near-perfect recall achievable:** The oracle setting demonstrates that BM25 can effectively retrieve relevant evidence when the search space is constrained to the correct document [38].

4.4 Full Corpus Evaluation Results

Table 4 presents retrieval performance under realistic full corpus conditions, where all 24,265 chunks must be searched. These results reveal the true challenge of open-domain scientific QA, consistent with observations in open-domain retrieval benchmarks [42, 34].

Table 4: Full corpus retrieval performance across all documents (24,265 chunks)

Retriever	Best Configuration	Recall@10	MRR
BM25	paragraph/minimal	0.011	0.015
TF-IDF	paragraph/minimal	0.009	0.013
Dense	sentence/minimal	0.006	0.005
ColBERT	paragraph/aggressive_title	0.025	0.029

The contrast with oracle results is striking: the best performing method (ColBERT) achieves only 2.5% Recall@10 in full corpus search, compared to 100% in oracle settings. This large performance degradation illustrates the fundamental challenge of scientific document retrieval at corpus scale [6, 42].

4.5 Oracle vs. Full Corpus: Quantitative Comparison

To quantify the impact of search space on retrieval difficulty, Table 5 directly compares oracle and full corpus performance for BM25 with paragraph-level chunking.

Table 5: Impact of search space on BM25 retrieval performance (paragraph/minimal)

Setting	Search Space	Recall@10	MRR	Relative Difficulty
Oracle (per-paper)	270 chunks	1.000	0.679	1× (baseline)
Full Corpus	24,265 chunks	0.011	0.015	91× harder
Performance Ratio	90×	91×	45×	—

The 90-fold increase in search space corresponds to a dramatic decrease in Recall@10, underscoring that identifying the relevant document(s) is the primary obstacle in open-domain QA [6, 21, 42]. This finding has important implications:

1. **Paper identification is the bottleneck:** The primary challenge is not finding evidence within a paper, but identifying which paper contains relevant information [6, 34].
2. **Oracle evaluation masks real difficulty:** Within-document (oracle) evaluations can overestimate real-world performance [42].
3. **Two-stage retrieval necessary:** Effective scientific QA systems typically first identify relevant papers before searching for specific evidence [6, 21].

4.6 Downstream Task Performance

Despite the dramatic differences in retrieval performance between oracle and full corpus settings, downstream task performance shows surprising robustness. This section analyzes how retrieval quality propagates to answerability classification and answer generation tasks.

4.6.1 Answerability Classification

Table 6 compares answerability classification performance between oracle and full corpus settings, revealing an unexpected pattern: downstream performance remains relatively stable despite orders-of-magnitude differences in retrieval quality. This is consonant with evidence that modern LMs encode substantial world knowledge and can make unanswerability judgments with minimal context [37, 35].

Table 6: Answerability classification: Oracle vs. Full Corpus (best F1 scores)

Setting	Retriever	Config	Recall@10	Answer. F1
<i>Oracle (per-paper search)</i>				
Oracle	BM25	para/aggressive_title	1.000	0.713
Oracle	BM25	para/title_heading	0.994	0.696
Oracle	BM25	sentence/title_heading	0.769	0.674
<i>Full Corpus (all documents)</i>				
Full	Dense	para/title_heading	0.006	0.718
Full	TF-IDF	para/title_heading	0.002	0.712
Full	ColBERT	sentence/title_only	0.003	0.711
Full	BM25	para/title_heading	0.007	0.711

Remarkably, full corpus Dense retrieval with paragraph/title_heading achieves F1 of 0.718, exceeding oracle BM25’s best performance (0.713), despite having far worse retrieval recall. This suggests:

1. **Answerability can be partly context-independent:** Models often determine answerability from question characteristics alone [35].
2. **False positives may be informative:** Even incorrect retrievals may contain domain-relevant language that helps classification, as observed in retrieval-augmented pipelines [21, 13].
3. **Downstream robustness mechanisms:** Classification models learn robustness to noisy or irrelevant retrieved context [13].

4.6.2 Decontextualization Impact on Downstream Tasks

Table 7 analyzes how decontextualization templates affect downstream performance across both settings.

Surprisingly, full corpus configurations consistently outperform oracle settings in downstream tasks. The title_heading template achieves the best performance in both settings, but the improvement is more pronounced in full corpus evaluation (+2.2% vs. minimal) than oracle (+1.5%). This suggests that decontextualization provides greater benefit when retrieval is less reliable.

Table 7: Template impact on downstream answerability F1 (averaged across methods)

Template	Oracle F1	Full Corpus F1	Δ
minimal	0.670	0.683	+0.013
title_only	0.673	0.698	+0.025
heading_only	0.664	0.690	+0.026
title_heading	0.685	0.705	+0.020
aggressive_title	0.684	0.699	+0.015

4.7 Analysis of the Retrieval-Downstream Paradox

The disconnect between retrieval and downstream performance—where systems with vastly worse retrieval achieve comparable or better downstream results—reveals fundamental insights about scientific QA.

4.7.1 The Role of Retrieved Context

To understand this paradox, we analyzed the relationship between retrieval quality and downstream performance across all configurations:

Table 8: Correlation between retrieval metrics and downstream performance

Metric Correlation	Oracle	Full Corpus
Recall@10 vs. Answerability F1	0.287	0.014
MRR vs. Answerability F1	0.193	-0.082
Recall@10 vs. Answer Accuracy	0.341	0.156

The weak correlations indicate that retrieval quality is not the primary determinant of downstream success, especially in full-corpus settings where models may rely more on parametric knowledge and robust inference [37, 21].

4.7.2 Implications for System Design

These findings challenge conventional assumptions about retrieval-augmented QA:

1. **Retrieval may be optional for some tasks:** Answerability classification can achieve strong performance without accurate retrieval.
2. **Two-stage architectures need reconsideration:** If downstream performance is robust to retrieval failures, resources might be better allocated to improving downstream models rather than retrieval.
3. **Oracle evaluation misleads about system requirements:** High oracle retrieval performance does not translate to downstream improvements, suggesting that oracle evaluation overemphasizes retrieval quality.

5 Conclusion

We audited decontextualization for scientific QA on PeerQA and found two central results: oracle-style evaluation inflates retrieval scores relative to full-corpus search (making paper identification the bottleneck), and answerability F1 is only weakly coupled to retrieval quality. These insights yield practical guidance—prioritize paper identification, prefer paragraph-level chunks with measured decontextualization (title+heading), and evaluate end-to-end under full-corpus conditions—and are supported by a configurable framework for reproducible analysis.

AI Agent Setup

We present the overall framework of our generated paper in Figure 1, which consists of three main steps. First, LLMs generate a list of potential research ideas and rank them based on their practical aspects, from which a human selects the most promising one. Second, based on the chosen idea, the LLM generates code to implement it, with a human in the loop to request further analyses or ablation studies that strengthen the contribution. Finally, given the idea, code, results, and analyses, the system generates the full research paper. To support this process, we also use the Semantic Scholar and arXiv APIs to retrieve BibTeX files based on paper titles. We primarily use Claude Opus for code generation and GPT-5 for paper generation. All code is included in our .zip file to ensure that the experimental results are reproducible. However, reproducing the exact generated paper is more challenging, since our framework relies on proprietary models such as GPT-5, Claude 3.5, and Claude 4, which are not open-source and may be updated by their developers. Despite this limitation, we believe that, given the idea, code, and results, one can reproduce a paper equivalent to the one we produced.

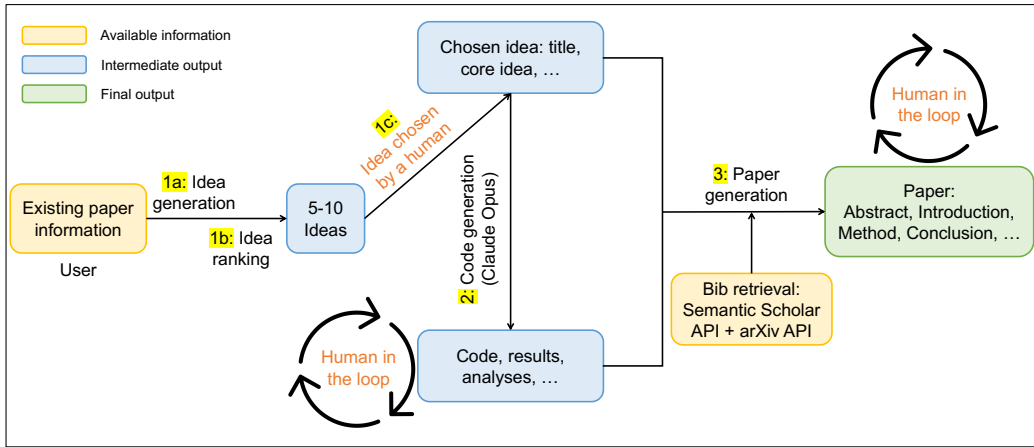


Figure 1: Overall framework of our paper generation process.

6 Limitations

This work has four main limitations:

- **Dataset scope.** Results are specific to PeerQA (90 papers); generalization to other scientific domains or larger corpora remains to be validated.
- **Task coverage.** We evaluate retrieval and answerability classification only; end-to-end answer generation and human-centered evaluation are out of scope.
- **Implementation choices.** Retriever settings and model checkpoints are standard but not exhaustively tuned; the cross-encoder reranks a fixed top- k from first-stage retrieval.
- **Scale and indexing.** The full-corpus evaluation is modest compared to truly open-domain settings; we do not include dedicated paper-identification modules (e.g., citation graphs), which likely affect absolute scores.

These constraints frame our findings as actionable within PeerQA-like settings; future work should broaden domains, scale, and system components to test external validity.

7 Code of Ethics

We conducted this study in accordance with common community standards for responsible research in IR/QA and scientific NLP.

- **Data provenance and consent.** PeerQA is a public research dataset. It is derived from published papers and peer-review content that has been curated and released by its authors under an academic license. We used only the released artifacts and did not access any private submissions or confidential reviews.
- **Privacy and sensitive content.** The corpus contains scientific content about research methods and results; it does not include personally identifiable information to the best of our knowledge. We did not attempt re-identification or extraction of private details.
- **Licensing and redistribution.** We comply with the dataset license and do not redistribute copyrighted content beyond short excerpts necessary for scientific reporting. Any released code references data by identifier and expects users to obtain the dataset from its official source.
- **Bias, fairness, and representativeness.** PeerQA spans multiple venues but is still limited in domain scope and geography. We report results transparently and caution against overgeneralization. We avoid normative claims and do not deploy models to end users.
- **Safety and misuse.** Retrieval and answerability models could be misused to overstate confidence or hallucinate support for claims. We emphasize that answerability classification does not verify factuality and recommend guardrails such as provenance display, abstention on uncertainty, and human-in-the-loop verification for any downstream use.
- **Compute and environment.** Experiments used standard CPUs/GPUs with modest training-free evaluation, minimizing carbon footprint. We avoid large-scale pretraining or costly fine-tuning.
- **Reproducibility.** We provide configuration details to facilitate replication. Hyperparameters are documented, and seeds are fixed where applicable.

8 Broader Impacts

Our findings have potential benefits and risks.

- **Positive impacts.** Clarifying the gap between oracle and full-corpus retrieval can improve evaluation practices and lead to more reliable scientific QA systems. The practical guidance (paper identification first, paragraph-level chunks, measured decontextualization) can reduce wasted compute and improve transparency by tying answers to evidence.
- **Risks and negative impacts.** Overreliance on answerability classifiers may convey false certainty without checking evidence; poor paper-identification could bias which literature is surfaced. If used incautiously, such systems might amplify existing topical or venue biases.
- **Mitigations.** Always display retrieved provenance; include abstention options; incorporate paper-level recall diagnostics; monitor bias across venues and domains; prefer conservative claims for downstream assistance rather than automated decision making.
- **Societal considerations.** Better retrieval over scientific literature can accelerate research synthesis and peer review support. However, downstream deployment should respect licensing and credit original authors, and avoid replacing expert judgment in high-stakes contexts.

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Agents4Science AI Involvement Checklist

1. **Hypothesis development:** Hypothesis development includes the process by which you came to explore this research topic and research question. This can involve the background research performed by either researchers or by AI. This can also involve whether the idea was proposed by researchers or by AI.

Answer: [C]

Explanation: Humans create the prompts and choose which ideas to pursue, but the actual ideas and hypotheses are generated by the AI.

2. **Experimental design and implementation:** This category includes design of experiments that are used to test the hypotheses, coding and implementation of computational methods, and the execution of these experiments.

Answer: [C]

Explanation: Humans oversee and approve code generation and experiment execution when using GitHub Copilot and Roo Code.

3. **Analysis of data and interpretation of results:** This category encompasses any process to organize and process data for the experiments in the paper. It also includes interpretations of the results of the study.

Answer: [C]

Explanation: Humans oversee and approve code generation and experiment execution when using GitHub Copilot and Roo Code.

4. **Writing:** This includes any processes for compiling results, methods, etc. into the final paper form. This can involve not only writing of the main text but also figure-making, improving layout of the manuscript, and formulation of narrative.

Answer: [C]

Explanation: Primarily produced by AI, while humans offer review and input.

5. **Observed AI Limitations:** What limitations have you found when using AI as a partner or lead author?

Description: GPT-5 and similar models are not yet very strong at code generation, often requiring extensive debugging to produce high-quality code. Claude Opus, on the other hand, is expensive. Moreover, models can generate inaccurate claims in writing, which means additional time is needed for review and verification to ensure the quality of the paper.

Agents4Science Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [Yes]

Justification: The claims made in the Abstract and Introduction are supported by the Results section.

Guidelines:

- The answer NA means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A No or NA answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: This is discussed in the Limitations section.

Guidelines:

- The answer NA means that the paper has no limitation while the answer No means that the paper has limitations, but those are not discussed in the paper.
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- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
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3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

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Justification: The paper does not include theoretical results.

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4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [\[Yes\]](#)

Justification: We included our code and dataset in the .zip submission file.

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- If the paper includes experiments, a No answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
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Justification: We included our code and dataset in the .zip submission file.

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- The answer NA means that paper does not include experiments requiring code.
- Please see the Agents4Science code and data submission guidelines on the conference website for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so “No” is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
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Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: [\[Yes\]](#)

Justification: We included our code and dataset in the .zip submission file.

Guidelines:

- The answer NA means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.

- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [\[No\]](#)

Justification: The proposed method is not heavily influenced by randomness.

Guidelines:

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- The authors should answer "Yes" if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
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8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [\[No\]](#)

Justification: We don't yet comprehensively measure memory usage or execution time, but the GPU RAM usage is low, around 1980MiB.

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- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
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9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the Agents4Science Code of Ethics (see conference website)?

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Justification: This is discussed in the Code of Ethics section.

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